

**Project report on
JUDICIARY INFORMATION SYSTEM**

**Project report submitted in partial fulfilment of the requirement for Software
Engineering Lab**

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2025-2026

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES (AP IIIT)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING 2025-2026



CERTIFICATE

This is to certify that the project report entitled “**JIS TOOL**” being submitted by **B.Glory, G.Niharika, B.Tejasree, M.Sneha** under my guidance and supervision and is submitted to **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING** in partial fulfilment of requirements for SoftwareEngineering Lab during semester – I of academic year 2025 - 2026 and it has been found worthy of Acceptance According to the requirements of the Department.

Signature of Internal Guide

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TABLE OF CONTENTS

1. WBS Document

- 1.1 About the Company
- 1.2 Project Name
- 1.3 Objectives / Aim of the Project
- 1.4 Users of the Product
- 1.5 Technologies Used
- 1.6 Work Assignment to Team
- 1.7 Work Breakdown Structure (WBS)

Pages: 5-7

2. Story Board

- 2.1 Introduction
- 2.2 Storyboard
- 2.3 Conclusion

Pages: 8-15

3. User Requirement Document (URD)

- 3.1 Introduction
- 3.2 Purpose
- 3.3 Intended Audience
- 3.4 Stakeholders
- 3.5 Product Vision
- 3.6 Technologies
- 3.7 System in Context
- 3.8 User Characteristics
- 3.9 Constraints
- 3.10 System-wide Requirements
- 3.11 Use Case Overview
- 3.12 Use Case Specifications
- 3.13 Non-Functional Requirements

4. Figures:

- Context Diagram – Page 19
- Use Case Diagram – Page 25
- ER diagram-page 26

Pages: 16-38

5.Website

- 4.1 Frontend Pages (HTML/JSP)
- 4.2 Backend Structure (Servlets)
- 4.3 Database (JDBC)

Pages: 39-52

WBS DOCUMENT

About Company

IDEA ARCHITECTS is developing the Judiciary Information System (JIS) for the Attorney General's office. Our mission is to create a robust platform that streamlines court case management, provides easy access to past court cases for lawyers and judges, and supports efficient judicial processes.

Project Name:-The Judiciary Information System (JIS) aims to:

- **Streamline Court Case Management:** Facilitate the entry and tracking of court case details including defendant information, crime type, dates, locations, and arresting officer details.
- **Automate Case Identification:** Generate unique case identification numbers (CINs) for each case.
- **Efficient Hearing Scheduling:** Assist registrars in assigning hearing dates by displaying vacant slots on working days.
- **Manage Case Progress:** Record reasons for adjournments, new hearing dates, and summaries of court proceedings.
- **Archive Case Information:** Maintain detailed records of closed cases, including judgment summaries, for future reference.
- **Provide Access to Judicial History:** Allow judges and lawyers to browse past court cases for guidance and preparation, with a mechanism to charge lawyers for access.
- **Support Querying and Reporting:** Enable court registrars to query pending cases, resolved cases within a period, upcoming hearings, and the status of any particular case.
- **Enable Keyword Search:** Allow searching of past court cases using keywords.
- **Manage User Accounts:** Provide registrars with the ability to create and delete login accounts for different users (judges, lawyers, etc.) and track lawyer case views for billing purposes.

Users of the Product:

- Court Registrars
- Judges
- Lawyers
- Police
- Attorneys General Office Staff

Technologies:

- **Frontend:** JSP (Java Server Pages)
- **Backend:** Servlets
- **Database:** JDBC (Java Database Connectivity)
- **Styling:** Tailwind CSS
- **Deployment/Cloud:** Google Cloud Platform (GCP) (or) Vercel

Work Assignment to Team Idea Architects

Work Name	Assigned Person
Requirements Gathering	Entire Team
Interface	Sneha
Story board	Tejasree, Niharika, Sneha
WBS	Glory, Tejasree, Niharika
URD	Entire Team
SRS	Entire Team
Risk Analysis	Entire Team
Front end Design (HTML pages)	Niharika, Glory
Backend Design (Database)	Tejasree, Sneha
Code Integration	Entire Team
Testing	Entire Team
Final Demo	Entire Team

Work Break Down Structure for JUDICIARY INFORMATION SYSTEM:-

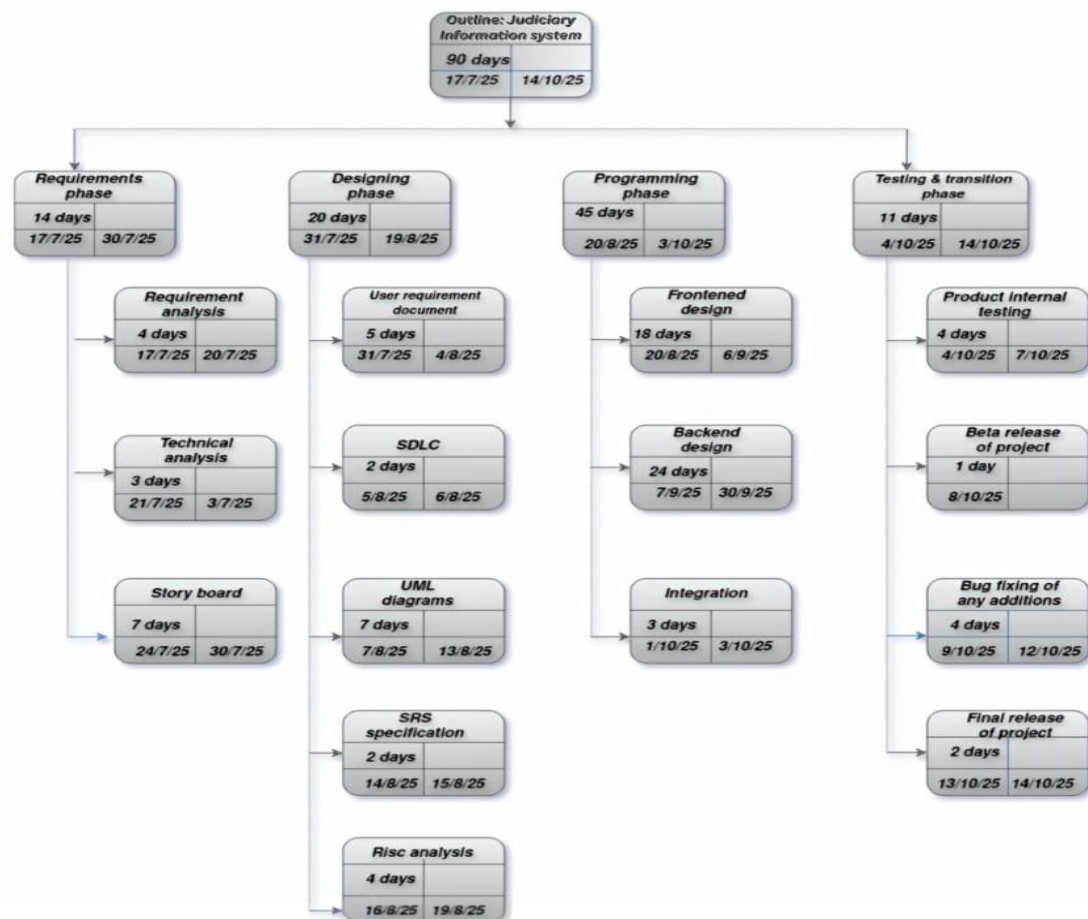


Table of Contents

STORY BOARD

Traditional Record Maintenance	01
The Record Lost	02
Confusion in Court	03
Case Delays and Restart	04
A Moment of Realization	05
Seeking a Solution	06
Designing the JIS	07
A Digital Transformation	08
How it WORKS?	09

1

VISION



"To transform the traditional, fragile, and manual system of court case management into a secure, efficient, and searchable digital solution that ensures timely justice and complete accountability. The Judiciary Information System (JIS) is envisioned as a centralized platform that empowers registrars, judges, and lawyers to access, manage, and retrieve case data reliably – preventing data loss, minimizing delays, and improving the overall efficiency of the judicial process."

2

USERS

1
COURT REGISTER

2
LAWYER

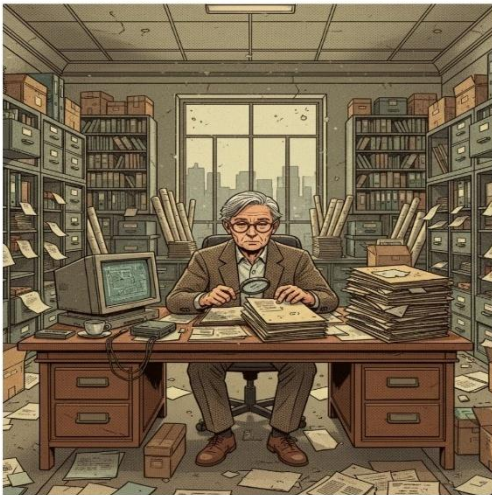
3
JUDGE

4
PUBLIC PROSECUTOR

5
POLICE



3



Traditional Record Maintenance

Court case details were maintained manually by the registrar in a traditional paper-based ledger system. This register contained all information about the cases — defendants, hearing dates, charges, and progress. Everything depended solely on this single, physical book. There was no backup, and any mistake or loss could severely impact the court's functioning

4

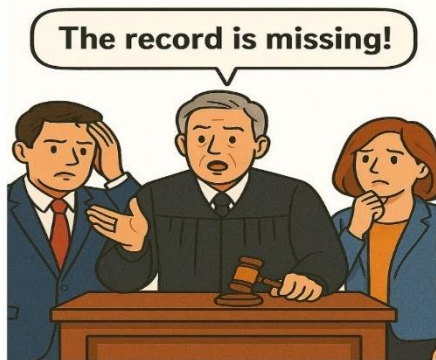
The Record lost!

One unfortunate morning, the registrar discovered that the book containing all court case records was missing. All case schedules, judge assignments, and hearing dates vanished with it. The court was left without any official reference or documentation to proceed with the hearings



5

Confusion in Court



As the court day began, judges and lawyers arrived expecting updates and schedules. With no records available, the registrar was unable to respond to their queries. Chaos erupted in the courtroom. No one knew which case was scheduled when, or who the involved parties were. The judicial process was thrown into disarray.

6

Case Delays and Restart



Due to the missing records, many court cases had to be postponed or even restarted from the beginning. Key details about witnesses, hearing dates, and evidence were no longer accessible. The delay in justice not only affected victims but also allowed certain cases to lose momentum and integrity.

7

A Moment of Realization



At the end of the day, the registrar reflected on the disaster. He realized that relying on a manual record system was no longer practical in a modern court setup. A single point of failure had resulted in the breakdown of the court's workflow. He knew a digital transformation was necessary.

8

Seeking a Solution

Due to the missing records, many court cases had to be postponed or even restarted from the beginning. Key details about witnesses, hearing dates, and evidence were no longer accessible. The delay in justice not only affected victims but also allowed certain cases to lose momentum and integrity.



9

Designing the JIS



The software team began designing the Judiciary Information System (JIS) — a secure , structured , and efficient digital case management solution. The system was tailored to store every detail: case ID, defendant info, hearing dates, judgments, and summaries — all in one centralized digital platform.

10

A Digital Transformation



Now, with JIS implemented, the court operates smoothly. Judges can access old judgments instantly, lawyers can refer to past cases (with tracking), and registrars can manage case data reliably. The risk of record loss is eliminated, ensuring a faster and more transparent judicial process.

11

HOW IT WORKS???

POLICE: ADDING CASE DETAILS



The officer opens the “Add Case Details” page and fills in the following information.

- Defendant’s Name
- Defendant’s address
- Date of Arrest
- Arresting Officer Name
- Crime Type(select from Drop down list)
- Place of Crime

once entered,the officer clicks Submit,and the system validates all field.

Purpose:To demonstrate the data entry flow of the new functionality.

12

CIN generation

After submission, the system automatically generates a unique Case Identification Number (CIN) and displays a confirmation message:

“Case Registered Successfully. CIN: JIS-C2025-0548”

The system saves all details securely in the database, linking the FIR and crime data for future use by registrars, judges, and lawyers.

Purpose:

To highlight automation and database integration for faster processing

13

REGISTRAR: Registrar Login



The Registrar opens the Judiciary Information System (JIS) portal and logs in with official credentials.

Once verified, the system opens the Registrar Dashboard, showing key modules:

- Register New Case
- Assign case to lawyers and judges
- Schedule Hearings
- View Case Lists(Pending,Resolved,Upcoming)

Purpose:To depict secure access and role based dashboard visibility for the Registrar

14

Registrar: Schedule /Update Hearing

The Registrar opens the Judiciary Information System (JIS) portal and logs in with official credentials.

Once verified, the system opens the Registrar Dashboard, showing key modules:

- Create/Remove Accounts
- Register New Case
- Schedule Hearings
- View case lists(Pending,Resolved and upcoming)

Purpose:

To depict secure access and role-based dashboard visibility for the Registrar.



15

REGISTRAR: Query and Reports



The Registrar can:

- View pending cases,Resolved cases and upcoming hearings
- search by CIN or date range

Purpose:To demonstrate the data retrieval and reporting capabilities for judicial administration

16

JUDGE: Judge Login

The Judge accesses the JIS portal and logs in with verified credentials.

The dashboard displays assigned cases, today's hearings, and pending reviews.

Purpose: To show personalized case dashboards for judges

View Assigned Cases

The Judge selects "Assigned Cases."

For each case, details such as defendant name, crime type, arrest date, lawyer, and public prosecutor appear. Judges can review past proceedings and evidence uploaded by the police.

Purpose: To highlight comprehensive case access before hearings.



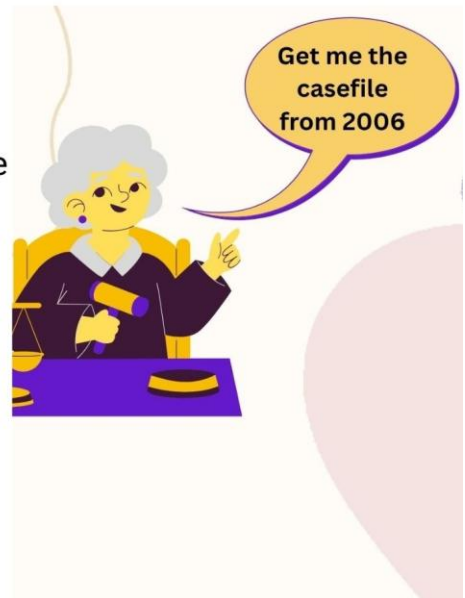
17

JUDGE: Browse Old Case Records

The Judge opens the "Reference Cases" module to browse past judgments for legal guidance.

The system allows search by keywords, CIN, crime type, or judgment date.

Purpose: To depict judicial research support through historical data access



LAWYER: Lawyer Refers to past case records by getting charged and view assigned cases

A Lawyer logs into the JIS Portal using secure credentials. The system opens the Lawyer Dashboard, showing options:

- Search Case by Case ID (CIN)
- Browse Past Court Cases
- View Assigned Cases
- View Payment History

The Lawyer searches past judgments using CIN to study prior rulings for case preparation.

Purpose: To highlight how JIS supports legal research and argument preparation by providing digital access to old case data.



User Requirement Document

~Judiciary Information System

Team: IDEA ARCHITECTS

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Version 1.1

Version History

Version no	Date	Changed by	Changes made
V1.1	2025-09-04	G. Niharika, B. Glory, B.Tejasree, M. Sneha	Initial draft of the requirements document, including system overview, user roles, core

			functionalities, and a proposed technology stack.
V1.2	2025-09-10	G. Niharika, B. Glory	Added details on the specific technologies, including JSP, Servlets, and a Google Cloud deployment plan.
V1.3	2025-09-11	B. Tejasree, M. Sneha	Refined the user role descriptions and clarified the system's ability to manage user accounts and track lawyer billing.

Introduction

This document details the requirements for the Judiciary Information System (JIS). The JIS is a tool requested by the Attorney General's Office to help manage court cases and provide accessible records for legal professionals.

Purpose

The primary purpose of this document is to outline the essential requirements needed to develop and implement the Judiciary Information System. It focuses on the system's key features, vision, scope, and overall functionality to provide an effective platform for managing court cases online. The JIS is intended to provide a centralized and secure solution for court case management and information retrieval.

Intended Audience

The intended audience for this system is comprised of various stakeholders who will use the tool for their daily tasks:

- **Police:** Responsible for data entry
- **Court Registrars:** case management, and user account administration.
- **Judges:** Users who need to access and reference past court cases for guidance.
- **Lawyers:** Users who require access to historical cases for legal research, subject to a per-view charge.

Stakeholders

- **Client:** The Attorney General's Office.
- **Users:** Police, Court Registrars, Judges, and Lawyers.

Product Vision

Vision Statement: The product vision is to develop a user-friendly and easily accessible Judiciary Information System that streamlines court case management, provides secure and efficient online solutions for all users, and maintains a comprehensive archive of legal proceedings.

Technologies

- **Front-End:** JSP (Java Server Pages)
- **Back-End:** Servlets
- **Database:** JDBC (Java Database Connectivity)
- **Styling:** Tailwind CSS
- **Deployment/Cloud:** Google Cloud Platform (GCP)

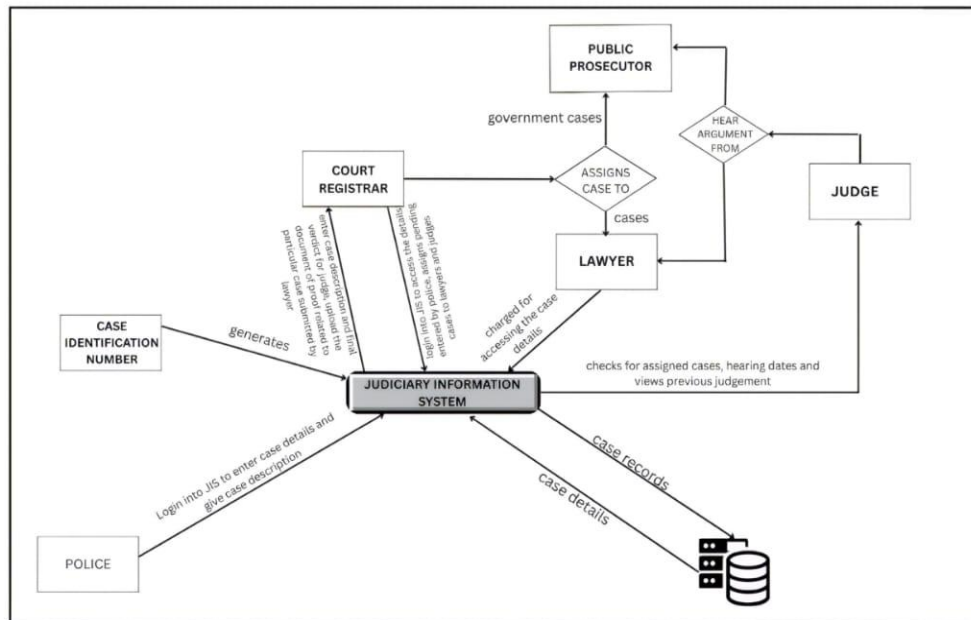
System in Context

The JIS is a multi-user system designed to manage the entire lifecycle of court cases. It provides a secure environment for different user roles with varying levels of access.

The system will:

- Accept new case requests from a court police, generating a unique Case Identification Number (CIN) for each.
- Allow the court registrar to enter all relevant case details, including defendant information, crime type, arresting officer details, and more.
- Provide a scheduling feature for the registrar to find and assign hearing dates, displaying vacant slots on any given working day.
- Enable registrars to update the status of a help request (e.g., adjourned, hearing takes place) and record a summary of proceedings or judgment.
- Provide a clear view of the status of any case to the registrar, judges, and lawyers.
- Allow the registrar to create and delete user accounts for judges and lawyers.
- Track the number of court cases a lawyer views for billing purposes.
- Facilitate browsing and searching of past court cases for both judges and lawyers, with appropriate access controls.

CONTEXT DIAGRAM



USER CHARACTERISTICS:

- **Court Registrar:** The registrar is an administrative user proficient in data entry and familiar with court procedures. They require an intuitive and efficient system for daily case management.
- **Judges:** Judges are legal experts who need a powerful search and reference tool. They are familiar with legal research and require a system that provides fast, accurate access to historical case data.
- **Lawyers:** Lawyers are professional legal researchers who require secure, keyword-based access to historical cases for legal preparation. They are accustomed to online research and billing systems.

CONSTRAINTS:

- **User Roles:** The system must account for three distinct user roles: Court Registrar, Judges, and Lawyers.
- **Registrar Functionality:** The registrar must be able to handle all administrative tasks, including data entry, case scheduling, status updates, query generation (pending, resolved, upcoming cases), and user account management.
- **Judges & Lawyers Access:** Both groups require read-only access to historical cases for research.
- **Lawyer Billing:** The system must track and charge lawyers for each case they view.

- **Data Integrity:** Each case must have a unique, computer-generated Case Identification Number (CIN).
- **Tone:** The language used in all documentation should be professional and meaningful.
- **Brevity:** Summaries or short versions of content should be concise and to the point.

SYSTEM-WIDE REQUIREMENTS:

Actors

- **Court Registrar:** The administrative user responsible for managing case data, scheduling, and user accounts.
- **Judges:** Users who access historical case information for legal guidance.
- **Lawyers:** Users who access historical case information for case preparation, with their views being tracked for billing.

Events

1. The Registrar enters new court case data, and the system generates a unique Case Identification Number (CIN).
2. The Registrar requests and views vacant time slots to schedule a hearing.
3. The Registrar enters a reason for a case adjournment or a summary of a court proceeding.
4. The Registrar enters a judgment summary and closes a case.
5. The Registrar queries the system for pending, resolved, or upcoming cases, or for the status of a specific case by its CIN.
6. A Judge browses and views past court cases for reference.
7. A Lawyer logs in and views a past court case, which is then recorded for billing.
8. The Registrar creates or deletes login accounts for Judges and Lawyers.

The below table provides a set of user visible events that define the functionalities that are in JIS.

ACTOR	ACTION	OBJECT	FREQUENCY	ARRIVAL PATTERN	RESPONSE
police	Login	Login page	1/day	Episodic	Police dashboard gets open
police	Selects	Post case webpage link	1/day	Episodic	Got a form which is used to

					fill the case details
Police	Posts	Web form	1/day	Episodic	Case details stored and got CIN (case identification number)
Police	Selects	View case status link	1/day	Episodic	Got a webpage to enter CIN
Police	Enters	CIN	1/day	Episodic	Got webpage contains case status
Registrar	Login	Login page	1/day	Episodic	Registrar dashboard gets open
Registrar	Selects	View all case details	1/day	Episodic	Got all case details in webpage
Registrar	Checks	Available lawyers, judges, public prosecutor	1/day	Episodic	Got all available lawyers, judges, public prosecutor
Registrar	Assigns	Case to judge and lawyer (or)public prosecutor	1/day	Episodic	Case details updated with assigned members
Registrar	Selects	Upload Documents of Proof link	1/day	Episodic	Upload Documents of Proof for a particular case

Registrar	Selects	Updates hearing dates link	1/day	Episodic	Upload hearing dates for pending cases
Registrar	Selects	Update case status link	1/day	Episodic	Got webpage contains case status
Registrar	Edits	Update case status	1/day	Episodic	Case status is updated
Lawyer	Login	Login page	1/day	Episodic	Lawyer dashboard gets open
Lawyer	Selects	View all assigned cases	1/day	Episodic	Got webpage with all assigned cases
Lawyer	Selects	Recently assigned case	1/day	Episodic	Got all the case details
Lawyer	Selects	Access previous case	1/day	Episodic	Request payment
Lawyer	completes	payment	1/day	Episodic	Access previous case details
Public prosecutor	Login	Login page	1/day	Episodic	Lawyer dashboard gets open
Public prosecutor	Selects	View all assigned cases	1/day	Episodic	Got webpage with all assigned cases
Public prosecutor	Selects	Recently assigned case	1/day	Episodic	Got all the case details
Public prosecutor	Selects	Access previous case	1/day	Episodic	Got all the case details

Judge	Login	Login page	1/day	Episodic	Lawyer dashboard gets open
Judge	Selects	View all assigned cases	1/day	Episodic	Got webpage with all assigned cases
Judge	Selects	Recently assigned case	1/day	Episodic	Got all the case details
Judge	Selects	Access previous case	1/day	Episodic	Got all the case details
Judge	Selects	Post final verdict link	1/day	Episodic	Got a form
judge	Posts	Final verdict	1/day	Episodic	Final verdict added to case details

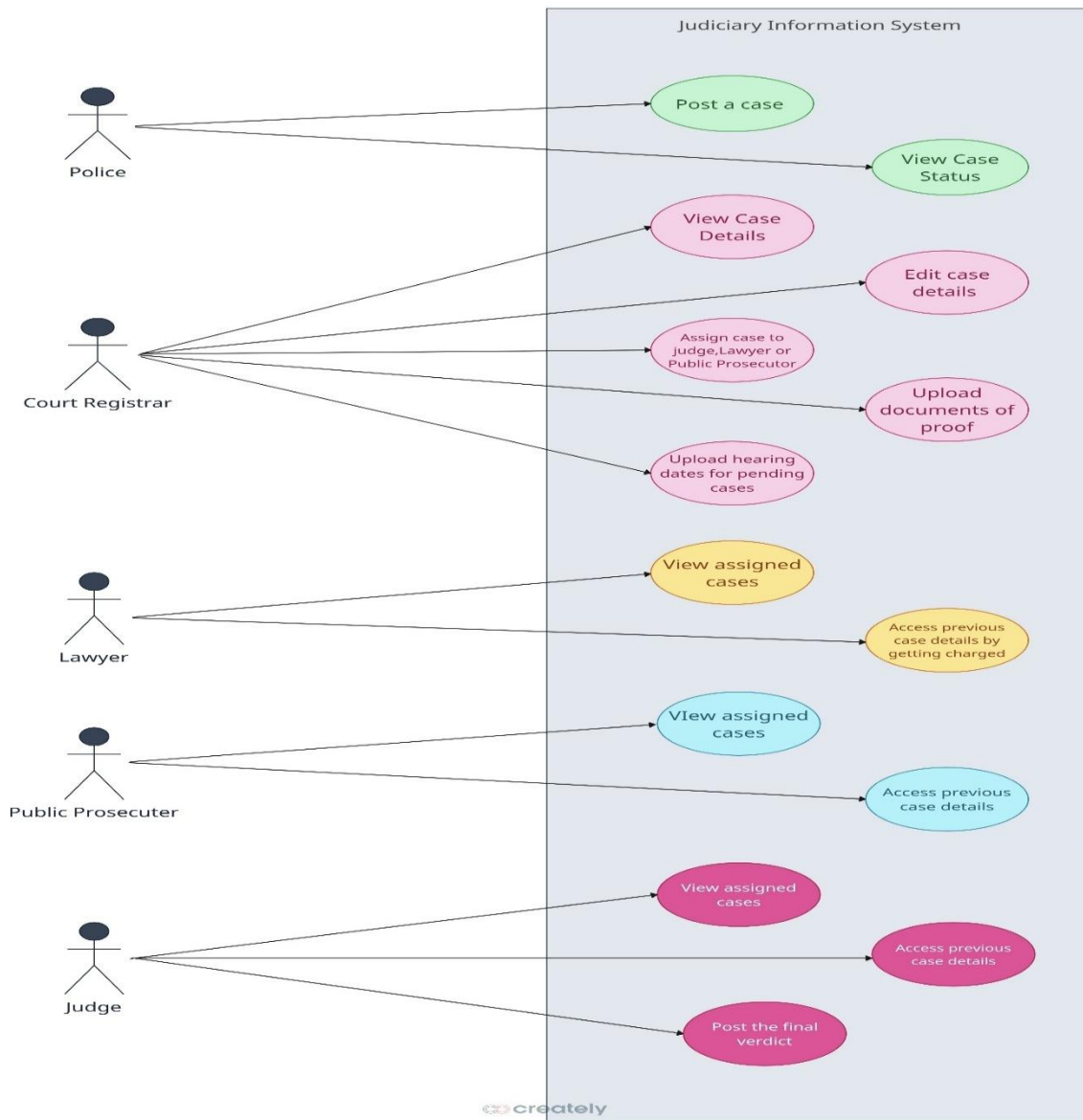
Functional Requirements :

Use-case Overview :

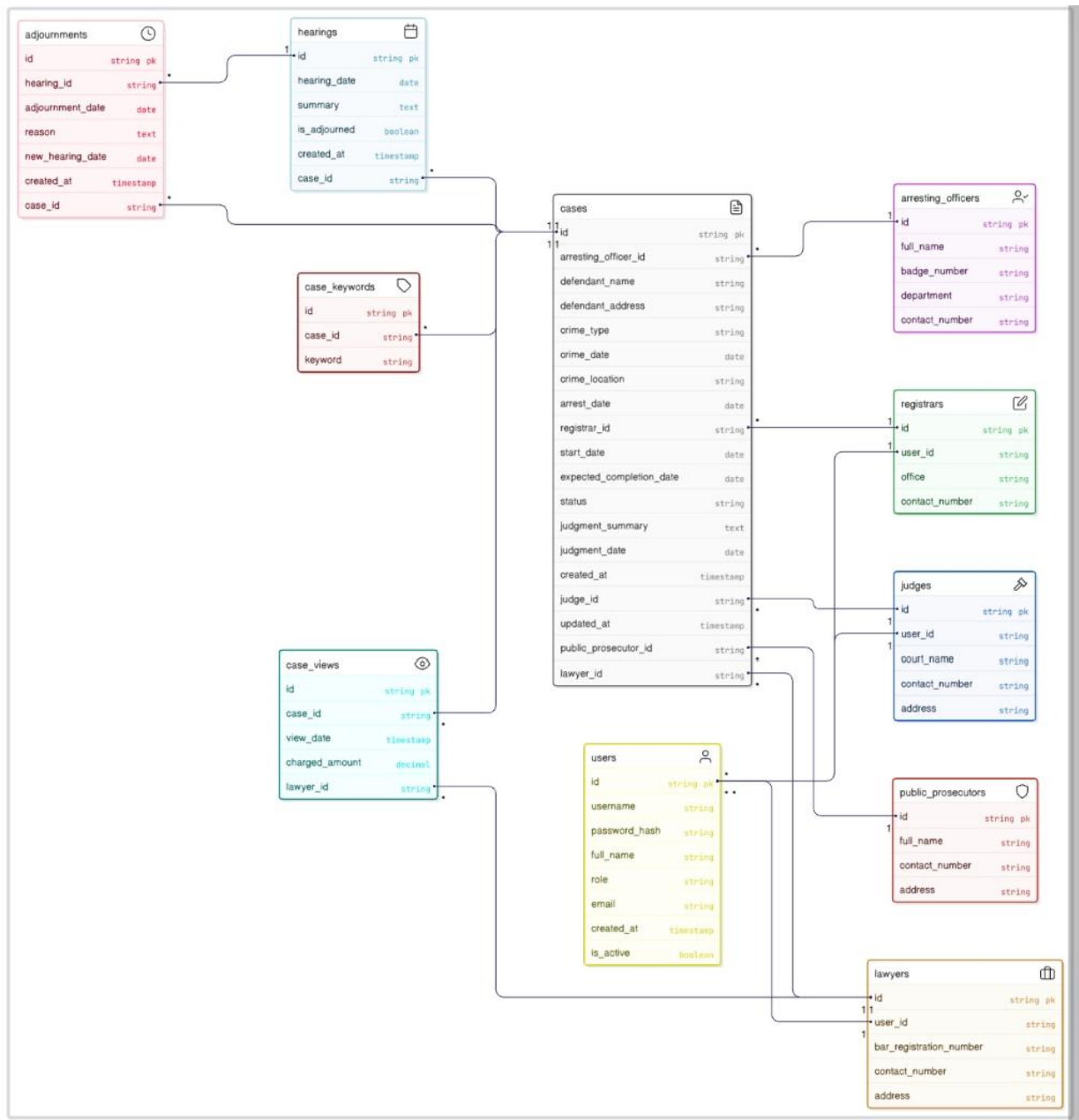
S.NO	USE CASE ID	USE CASE NAME	PRIORITY	STABILITY	VERIFIABILITY
1	UC-JIS-PC	Post case	High	stable	verifiable
2	UC-JIS-VCS	View case status	High	stable	verifiable
3	UC-JIS-VCD	View case details	High	stable	verifiable
4	UC-JIS-ECD	Edit case details	High	stable	verifiable

5	UC-JIS-AC	Assigns case	High	stable	verifiable
6	UC-JIS-UDP	Uploads documents of proof	High	stable	verifiable
7	UC-JIS-UHDPC	Updates hearing dates for pending cases	High	stable	verifiable
8	UC-JIS-VAC	Lawyer_ view assigned cases	High	stable	verifiable
9	UC-JIS-APC	Lawyer_ access previous cases	High	stable	verifiable
10	UC-JIS-VAC	Public prosecutor_ view assigned cases	High	stable	verifiable
11	UC-JIS-APC	Judge_ access previous cases	High	stable	verifiable
12	UC-JIS-VAC	Judge_ view assigned cases	High	stable	verifiable
13	UC-JIS-APC	Judge_ access previous cases	High	stable	verifiable
14	UC-JIS-PFV	Post final verdict	High	stable	verifiable

USE CASE DIAGRAM



ER DIAGRAM



USE-CASE SPECIFICATION:-

1.UC-JIS-PC: Post Cases

Use-Case ID	UC-JIS-PC
Use-Case Name	Post Case
Description	A Police submits a new court case, and the system generates a unique Case Identification Number (CIN).
Pre-conditions	The Police is logged in. The user has a new case to submit with all required details.
Success guarantee (post-conditions)	Case details are stored, a unique CIN is generated, and the user receives a confirmation with the CIN.
Main success scenario	The Police fills and submits a web form with case details, and the system provides a new CIN.
Frequency of occurrence	Episodic (1/ day)

2.UC-JIS-VCS: View Case Status

Use-Case ID	UC-JIS-VCS
Use-Case Name	View Case Status
Description	A Police user checks the current status of a submitted case using its CIN.
Pre-conditions	The Police user is logged in. The user has the unique CIN for the case they want to check.
Success guarantee (post-conditions)	The system provides the user with the current status of the requested case.
Main success scenario	The Police user enters a CIN on a webpage and receives the case status and details.
Frequency of occurrence	Episodic (1/ day)

3.UC-JIS-VCD: View Case Details

Use-Case ID	UC-JIS-VCD
Use-Case Name	View Case Details

Description	The Court Registrar views all the details of a specific case, including its status and assigned personnel.
Pre-conditions	The Court Registrar is logged in. The Registrar has access to the case identification number.
Success guarantee (post-conditions)	The Registrar can successfully view the full details of a particular case.
Main success scenario	The Registrar navigates to the "View all case details" section and selects a case from a list or searches by CIN.
Frequency of occurrence	Episodic (1/day)

4.UC-JIS-ECD: Edit Case Details

Use-Case ID	UC-JIS-ECD
Use-Case Name	Edit Case Details
Description	The Court Registrar updates existing case information, such as a judgment summary or an adjournment reason.
Pre-conditions	The Court Registrar is logged in. The Registrar has selected an existing case to edit.
Success guarantee (post-conditions)	The case details are successfully updated in the system.
Main success scenario	The Registrar selects a case to edit, modifies the details in a form, and submits the changes.
Frequency of occurrence	Episodic (1/day)

5.UC-JIS-AC: Assigns Case

Use-Case ID	UC-JIS-AC
Use-Case Name	Assigns Case
Description	The Court Registrar assigns an unassigned case to a Judge and a Lawyer or Public Prosecutor.
Pre-conditions	The Court Registrar is logged in. There is at least one new case available that has not been assigned.
Success guarantee	The case details are updated with the assigned legal professionals, who are then notified.

(post-conditions)	
Main success scenario	The Registrar selects a pending case, chooses the appropriate legal professionals from a list, and assigns them.
Frequency of occurrence	Episodic (1/ day)

6.UC-JIS-UDP: Upload Documents of Proof

Use-Case ID	UC-JIS-UDP
Use-Case Name	Uploads documents of proof
Description	The Court Registrar uploads supporting documents and evidence for a specific case.
Pre-conditions	The Court Registrar is logged in. There is an existing case to which documents need to be attached.
Success guarantee (post-conditions)	The documents are successfully uploaded and linked to the case record.
Main success scenario	The Registrar selects a case, uses a file upload interface, and attaches the documents to the case record.
Frequency of occurrence	Episodic (1/ day)

7.UC-JIS-UHDPC: Updates hearing dates for pending cases

Use-Case ID	UC-JIS-UHDPC
Use-Case Name	Updates hearing dates for pending cases
Description	The Court Registrar updates the hearing date for a pending case.
Pre-conditions	The Court Registrar is logged in. There is a pending case that requires an updated hearing date.
Success guarantee (post-conditions)	The hearing date for the specific case is successfully updated in the system.
Main success scenario	The Registrar selects a case, chooses a new vacant hearing date, and submits the update.
Frequency of occurrence	Episodic (1/ day)

8.UC-JIS-VAC: Lawyer_View Assigned Cases

Use-Case ID	UC-JIS-VAC
Use-Case Name	Lawyer_View Assigned Cases
Description	Lawyer view a list of all cases assigned to them.
Pre-conditions	The Lawyer is logged in.
Success guarantee (post-conditions)	The Lawyer can view a list of all cases currently assigned to them.
Main success scenario	The Lawyer logs in, selects "View all assigned cases," and the system displays their case list.
Frequency of occurrence	Episodic (1/ day)

9.UC-JIS-APC: Lawyer_Access Previous Cases

Use-Case ID	UC-JIS-APC
Use-Case Name	Lawyer_Access Previous Cases
Description	A Lawyer accesses historical court records for research. A Lawyer's access is tracked for billing.
Pre-conditions	The Lawyer is logged in.
Success guarantee (post-conditions)	The Lawyer can successfully view the details of past court cases. The system records the access for billing .
Main success scenario	The Lawyer navigates to "Access previous cases," searches for a case, and views the details.
Frequency of occurrence	Episodic (1/ day)

10.UC-JIS-VAC: Public Prosecutor_View Assigned Cases

Use-Case ID	UC-JIS-VAC
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Use-Case Name	Public prosecutor_ view assigned cases
Description	A Public Prosecutor views all the government cases that have been assigned to them.
Pre-conditions	The Public Prosecutor has successfully logged into the system.
Success guarantee (post-conditions)	The Public Prosecutor can view a list of all cases currently assigned to them.
Main success scenario	The Public Prosecutor logs in, selects "View all assigned cases," and the system displays their case list.
Frequency of occurrence	Episodic (1/day)

11.UC-JIS-APC: Public Prosecutor_Access Previous Cases

Use-Case ID	UC-JIS-APC
Use-Case Name	Public Prosecutor_ access previous cases
Description	A Public Prosecutor accesses historical court case records for research and reference purposes. This is a read-only action.
Pre-conditions	The Public Prosecutor has successfully logged into the system.
Success guarantee (post-conditions)	The Public Prosecutor can successfully view the details of past court cases.
Main success scenario	The Public Prosecutor navigates to "Access previous cases," searches for a case, and views the details.
Frequency of occurrence	Episodic (1/day)

12.UC-JIS-VAC: Judge_View Assigned Cases

Use-Case ID	UC-JIS-VAC
Use-Case Name	Judge_view assigned cases
Description	A Judge views a list of all court cases that have been assigned to them.
Pre-conditions	The Judge has successfully logged into the system.
Success guarantee	The Judge can view a list of all cases currently assigned to them.

(post-conditions)	
Main success scenario	The Judge logs in, selects "View all assigned cases," and the system displays their case list.
Frequency of occurrence	Episodic (1/ day)

13.UC-JIS-APC: Assigns Previous Cases

Use-Case ID	UC-JIS-APC
Use-Case Name	Judge_ access previous cases
Description	A Judge accesses historical court case records for research and reference purposes. This is a read-only action.
Pre-conditions	The Judge has successfully logged into the system.
Success guarantee (post-conditions)	The Judge can successfully view the details of past court cases.
Main success scenario	The Judge navigates to "Access previous cases," searches for a case, and views the details.
Frequency of occurrence	Episodic (1/ day)

14.UC-JIS-PFV: Post Final Verdict

Use-Case ID	UC-JIS-PFV
Use-Case Name	Post Final Verdict
Description	A Judge enters the final verdict and judgment summary, officially closing the case in the system.
Pre-conditions	The Judge is logged in. The case is ready for a verdict.
Success guarantee (post-conditions)	The final verdict and judgment summary are recorded, and the case status is updated to "Closed".
Main success scenario	The Judge selects a case, fills out a form with the verdict details, and submits to close the case.
Frequency of occurrence	Episodic (1/ day)

Software Requirements Specification (SRS) for Judiciary Information System

1. Introduction

1.1. Purpose

The primary purpose of this Software Requirements Specification (SRS) is to precisely define the functional and non-functional requirements for the **Judiciary Information System (JIS)**.

The **main objective of the JIS** is to modernize and streamline judicial operations for the Attorney General's office by providing a secure, efficient, and transparent digital platform. Specifically, it aims to:

- Efficiently handle active court cases from initial registration through judgment.
- Make past court cases easily and securely accessible to authorized lawyers and judges for reference and guidance.
- Automate data entry and management processes for court registrars.
- Provide tools for effective court scheduling and case tracking.
- Enable specific querying and reporting functionalities for key stakeholders.
- Implement a fair mechanism for cost recovery related to historical case access for lawyers.

1.2. Scope

The JIS will be a web-based application designed to streamline and manage various judicial processes as requested by the Attorney General's office.

In Scope:

- Detailed capture and management of court case information (defendant details, crime, dates, locations, officers).
- Generation of unique Case Identification Numbers (CINs).
- Dynamic management of hearing dates, including vacant slot display, adjournments, and recording of proceedings summaries.
- Recording of judgment summaries and case closure, with data retention for future reference.
- Management of presiding judge, public prosecutor, starting, and expected completion dates for trials.
- Secure, role-based access for judges, lawyers, and registrars.
- Tracking and charging mechanism for lawyers access to historical case details.
- Specific query functionalities for court registrars regarding pending, resolved, and upcoming cases, as well as individual case status.

Out of Scope:

- Direct integration with external financial transaction systems for automated billing.
- Offline access capabilities.
- Mobile-native applications (initially, it will be web-responsive).

1.3. Definitions, Acronyms, and Abbreviations

- **JIS:** Judiciary Information System
- **SRS:** Software Requirements Specification

- **CIN:** Case Identification Number - A unique identifier generated by the system for each court case.
- **Registrar:** Court Registrar/Administrator - The user role responsible for data entry, scheduling, and user account management.
- **Attorney General's Office:** The requesting authority and primary stakeholder for the JIS development.
- **HTTPS:** Hypertext Transfer Protocol Secure
- **Authentication:** Verifying user identity to grant access to the system.
- **Authorization:** Granting specific permissions and access levels to authenticated users based on their role.
- **Case:** A legal proceeding or dispute managed within the JIS.
- **Hearing:** A scheduled session where a court case is presented or discussed.
- **Adjournment:** The postponement or rescheduling of a court hearing.
- **Judgment:** The final decision or verdict in a court case.

1.4. References

- Storyboards for Judiciary Information System (Previous document/discussions)
- Relevant national/local judicial regulations and guidelines.

2. Overall Description

2.1. Product Perspective

The JIS is a standalone web application that will serve as the primary digital platform for managing judicial processes as specified by the Attorney General's office.

2.1.1. System Interfaces

- **Database Management System (DBMS):** The core system responsible for persistence, storage, and retrieval of all JIS operational and historical data.

2.1.2. User Interfaces

The system will feature a responsive web-based UI accessible via standard web browsers. It will be designed for intuitive navigation and ease of use, adhering to modern UI/UX principles. Key UI elements will include:

- Secure Login/Logout screens with distinct interfaces or dashboards based on user roles.
- Dedicated forms for court case data entry and updates.
- Interactive calendar view for scheduling and managing hearings.
- Search and listing screens for pending, resolved, and upcoming cases.
- Detailed case view screens with tabbed or sectioned navigation for comprehensive information (defendant, crime, proceedings, judgment).
- Document upload, viewing, and versioning interfaces.

2.1.3. Software Interfaces

The system will interact with:

Web Browsers: Chrome, Firefox, Edge, Safari (latest stable versions).

Operating Systems: Windows, macOS, Linux (for client-side access).

Technologies:

- **Frontend:** JSP (Java Server Pages)
- **Backend:** Servlets
- **Database:** JDBC (Java Database Connectivity)
- **Styling:** CSS
- **Deployment/Cloud:** Google Cloud Platform (GCP) **(or)** Vercel

2.1.4. Communications Interfaces

The system will communicate over **HTTPS** for secure data transmission. All network communications between clients and servers, and between different system components, will be encrypted to ensure data confidentiality and integrity.

2.1.5. Operations

The system will operate 24/7, with scheduled maintenance windows communicated in advance. It will require continuous monitoring for performance, security, and availability. Robust backup and recovery procedures will be in place to minimize data loss and ensure rapid system restoration in the event of failure.

2.2. Product Functions

The core functions of the JIS are designed to meet the specific needs outlined by the Attorney General's office:

- **Case Information Capture:** Detailed recording of all relevant case data upon initiation, including defendant details, crime specifics, dates, locations, and arresting officer.
- **Case Identification:** Automated generation and assignment of unique Case Identification Numbers (CINs) for each case.
- **Hearing and Scheduling Management:** Facilitating the assignment of hearing dates, displaying vacant slots, managing adjournments (with reasons), and recording summaries of court proceedings after each hearing.
- **Historical Case Access for Judges:** Providing judges with secure and unrestricted browsing capability for past court cases to aid in judgment guidance.
- **Historical Case Access for Lawyers (Chargeable):** Allowing lawyers to browse old cases via keyword search, with a system to track and charge for each access to case details.
- **Query and Reporting Capabilities:** Providing the registrar with specific functionalities to query and generate reports on:
 - Currently pending court cases (sorted by CIN, with specific details).
 - Cases scheduled for hearing on a particular date.
 - Status of any particular case by CIN.
- **Document Management:** Enabling the upload, storage, categorization, secure retrieval, and versioning of all case-related documents.

2.3. User Characteristics

- **Judges:** Highly privileged users needing efficient access to current and historical case details, orders, and comprehensive scheduling views. They require clear presentation of critical information for decision-making and guidance.

- **Lawyers/Attorneys:** Users focused on filing documents, tracking case progress, and accessing specific current case details. Critically, they require the ability to browse historical cases via keyword search, understanding that access to details will be tracked for billing purposes. They expect easy document submission and search.
- **Court Registrars/Administrators:** Users with high access levels, primarily responsible for accurate data entry for new and ongoing cases, managing hearing schedules, recording proceedings and judgments, and administrative tasks including creating and deleting user accounts. They require robust data entry forms, efficient administrative tools, and the ability to perform detailed queries and generate reports.
- **Police:** Police users are responsible for creating and submitting case reports, updating crime and arrest details, and uploading evidence as the first point of contact in the justice system. They require a simple, fast interface because their work is time-sensitive, often performed in the field, and demands accuracy. Their access is restricted to data entry and case tracking, without permission to modify court assignments or judge/lawyer decisions.

2.4. Constraints

- **Regulatory & Legal Compliance:** The JIS must rigorously comply with all applicable national and local judicial laws, court rules, data privacy regulations (e.g., GDPR, or equivalent local data protection acts), and confidentiality mandates regarding data access, retention, and disclosure.
- **Security:** Adherence to industry best practices for web application security (e.g., OWASP Top 10) is mandatory. This includes robust authentication, authorization, data encryption (in transit and at rest), and protection against common cyber threats. Special attention must be paid to securing lawyer billing information and preventing unauthorized access to historical case details.
- **Performance:** The system must handle concurrent users efficiently without significant degradation in response time. All pages should load within 3 seconds. Search queries for current and historical cases must return results within 2 seconds for expected data volumes (e.g., up to 100,000 cases). Document uploads (up to 50MB) should complete within 10 seconds under optimal network conditions.
- **Data Integrity:** Ensure the accuracy, consistency, reliability, and validity of all stored data, especially critical case details, hearing dates, and lawyer access logs for billing.
- **Technology Stack:** Preference for established, secure, and maintainable technologies. Consideration for open-source alternatives where feasible to optimize licensing costs without compromising security or performance.
- **Billing Logic:** The system must accurately track and count each instance a lawyer views the detailed content of a historical court case for future billing.

2.5. Assumptions and Dependencies

- **Internet Connectivity:** All users (judges, lawyers, registrars, public users) will have stable and sufficient internet access to operate the web-based system effectively.
- **Hardware Availability:** Necessary server infrastructure (computing, storage, networking) and client-side devices will be provisioned and maintained to support the system's operation.
- **User Training:** Adequate training and support will be provided to all judicial staff (judges, lawyers, registrars) to ensure effective system adoption and utilization.

- **Legal Framework for Billing:** A clear legal and administrative framework for charging lawyers for access to old cases will be established outside the scope of this software's development, providing the basis for the billing tracking functionality.

3. Specific Requirements

3.1. Functional Requirements

- **User Management:**

The system SHALL allow users (Registrars, Judges, Lawyers, Police) to register with unique credentials (username/email and password).

- The system SHALL authenticate users based on their provided credentials.
- The system SHALL support role-based access control (Registrar, Judge, Lawyer, Police) to restrict functionality and data view.

Case Information Management:

- For each new case, the system SHALL allow Police to enter case details and JIS automatically generate and assign a unique Case Identification Number (CIN).

The system SHALL allow the Registrar to create a new court case record.

- The system SHALL allow the Registrar to enter other mandatory case details: name of the presiding judge, name of the public prosecutor, the starting date of the trial, and the expected completion date of the trial.
- The system SHALL allow the Registrar to update any details of a currently active court case.
 - Upon completion of a court case, the system SHALL allow the Registrar to record a summary of the judgment and close the case.
 - The system SHALL retain all details of closed cases for future reference and historical access.

- **Hearing & Scheduling Management:**

- The system SHALL allow the Registrar to assign an initial date of hearing for each new case.
 - When assigning a hearing date, the system SHALL display vacant slots on any working day to the Registrar, considering court availability and judge's schedule.
 - The system SHALL allow the Registrar to enter the reason for adjournment and assign a new hearing date when a case is adjourned.
 - After a hearing takes place for a case, the system SHALL allow the Registrar to enter a summary of the court proceedings.

After a hearing (whether proceedings occurred or not), the system SHALL allow the Registrar to assign a new hearing date if the case is not closed.

The system SHALL provide a calendar view that displays all scheduled hearings for a given date, week, or month.

- **Case Querying & Reporting (Registrar):**

The system SHALL allow the Registrar to query all currently pending court cases.

- In response to the pending cases query, the system SHALL print out the cases sorted by CIN.
- The system SHALL allow the Registrar to query cases that have been resolved over any given period (date range).
- The system SHALL allow the Registrar to query cases that are coming up for hearing on a particular date.

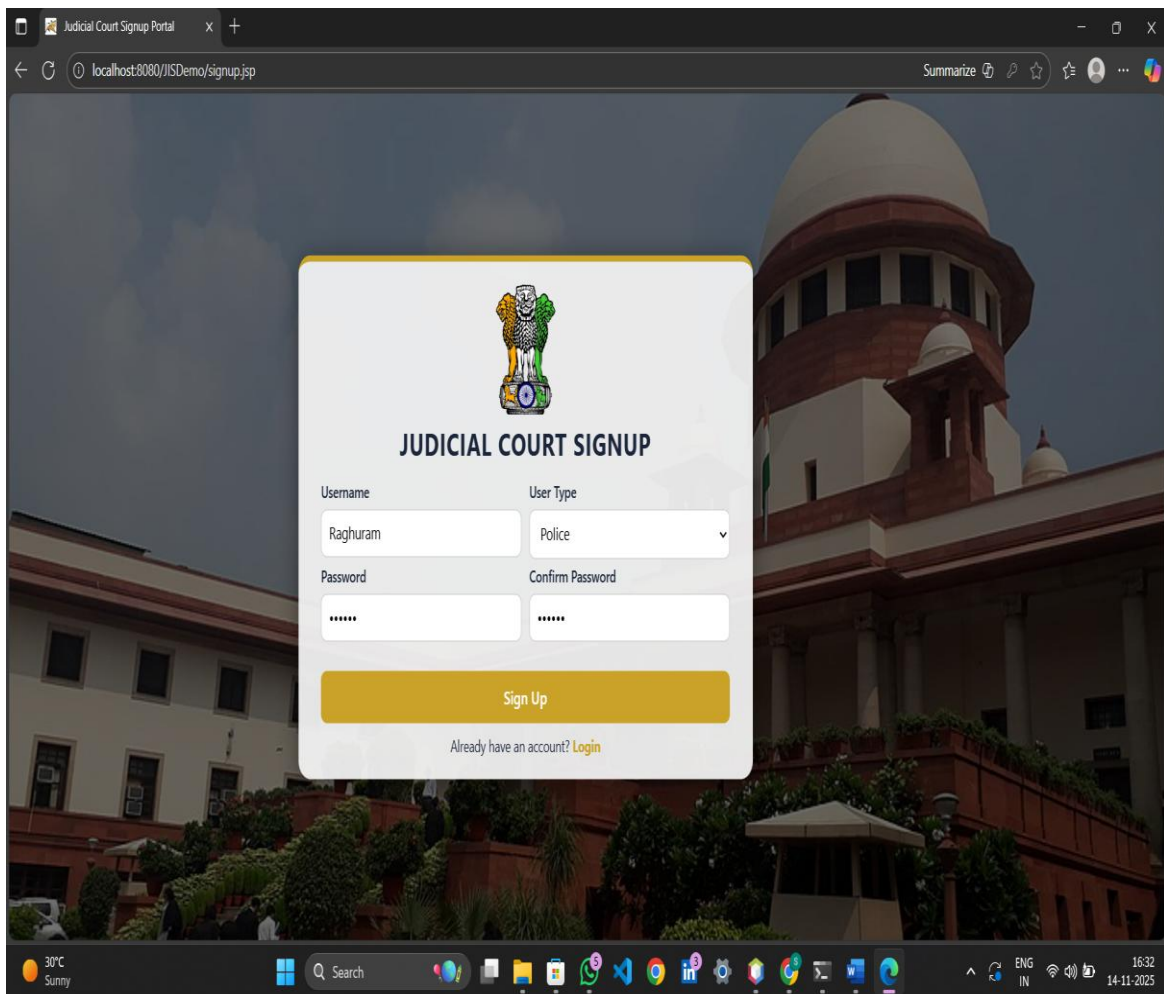
- The system SHALL allow the Registrar to query the current status of any particular case, identified by its CIN.
- **Historical Case Access & Billing:**
 - The system SHALL allow Judges to browse through the old (closed) cases for guidance on their judgment, with unrestricted access to details.
 - The system SHALL allow Lawyers to browse through old (closed) cases to prepare their lines of arguments.
 - The system SHALL charge Lawyers for each time they view the details of a closed court case.
 - The system SHALL provide the Registrar with a report or interface to view the usage count for each lawyer's historical case access.

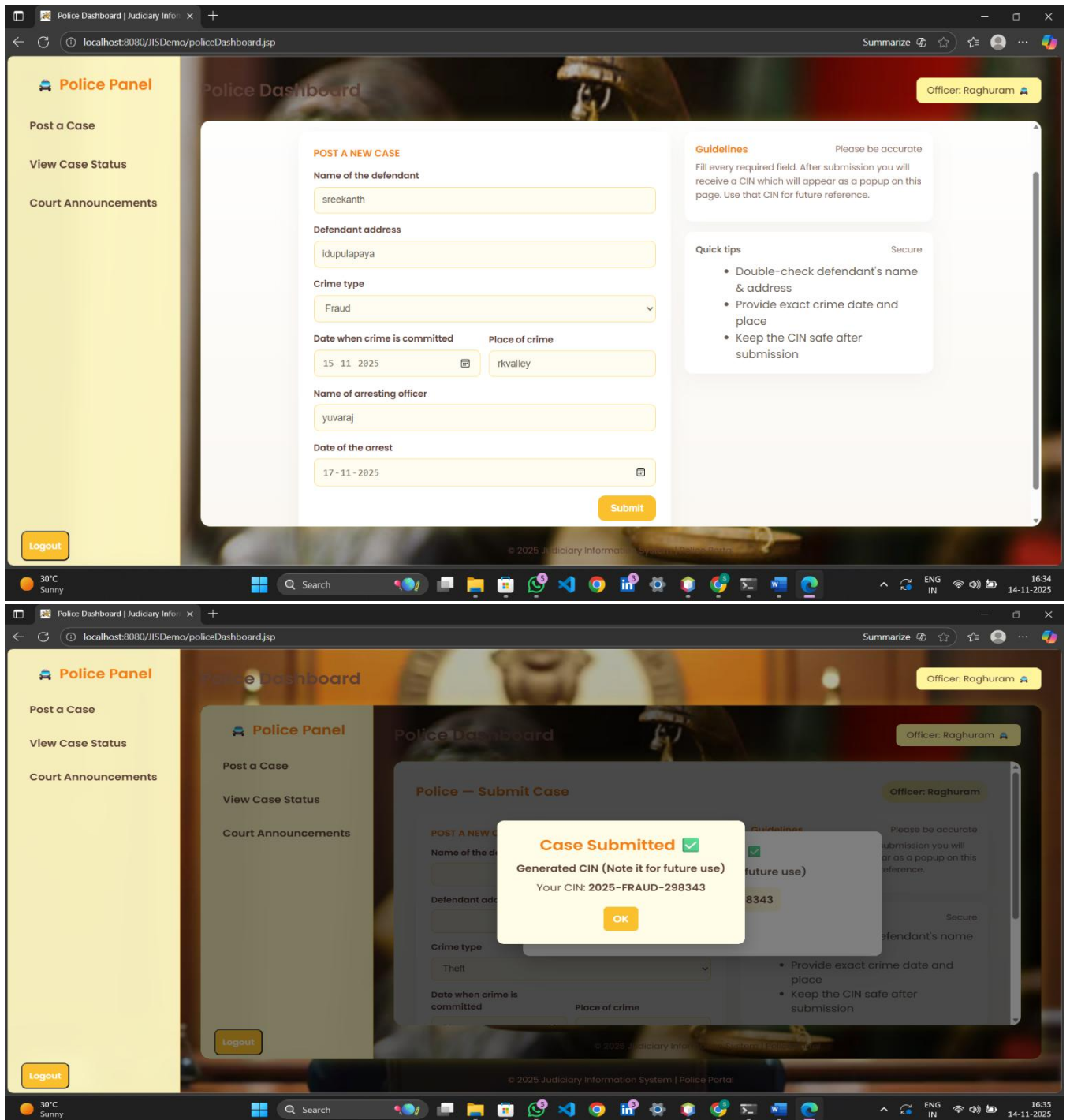
3.2. Non-Functional Requirements

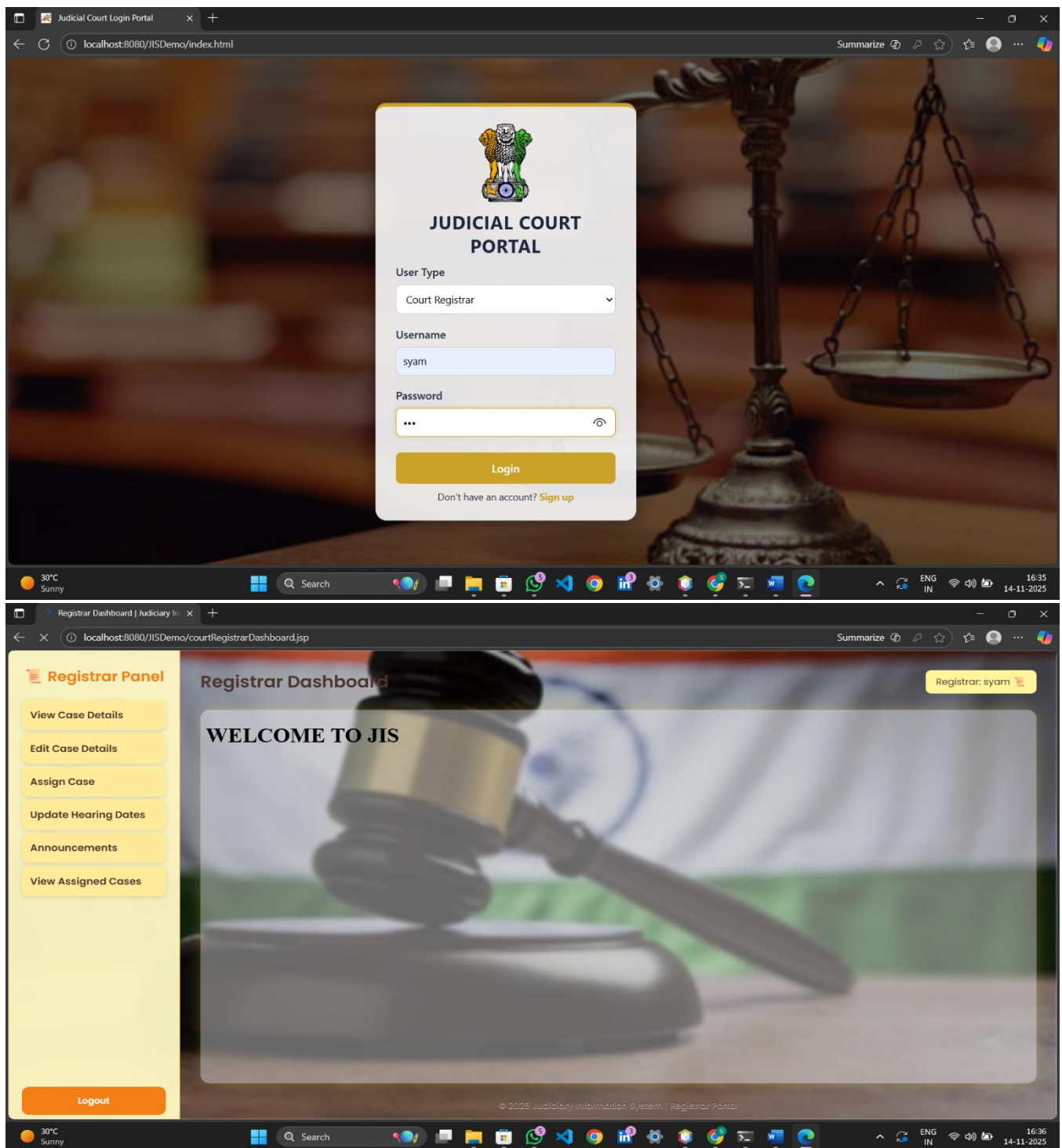
- **Performance:**
 - The system SHALL load all application pages within 3 seconds under normal load (up to 100 concurrent users).
 - Search queries (for current and historical cases, including keyword searches) SHALL return relevant results within 2 seconds for a database containing up to 500,000 cases.
 - Document uploads (up to 50MB) SHALL complete within 10 seconds under optimal network conditions.
- **Security:**
 - The system SHALL implement robust role-based access control mechanisms to ensure users can only access functionalities and data commensurate with their assigned roles.
 - The system SHALL log all critical security events, including but not limited to: failed login attempts, successful logins, unauthorized access attempts, and data modification events.
 - The tracking data for lawyer's paid views SHALL be securely stored and protected from unauthorized modification.
- **Reliability:**
 - The system SHALL have an overall uptime availability of 99.5% during operational hours, excluding scheduled maintenance windows.
 - The system SHALL handle unexpected errors gracefully, displaying informative and user-friendly error messages without exposing sensitive system details or crashing.
- **Usability:**
 - The system User Interface (UI) SHALL be intuitive, consistent, and easy to navigate for all user roles, minimizing the learning curve.
 - New users (Judges, Lawyers, Registrars, Police) SHALL be able to perform their primary tasks with minimal initial training (e.g., less than 2 hours of formal training).
 - The system SHALL provide clear and immediate feedback for all user actions, including success messages, validation errors, and loading indicators.
 - The system SHALL adhere to a consistent visual design and branding throughout, ensuring a professional and coherent user experience.
 - Forms for data entry shall be designed for efficiency, minimizing redundant input and guiding the user.
- **Portability:**

- The system SHALL be designed to be deployable on various cloud platforms (e.g., AWS, GCP, Azure) or on-premise infrastructure with minimal configuration changes.
- The client-side application SHALL be compatible with the latest two stable versions of major web browsers (Chrome, Firefox, Edge, Safari).
- **Accessibility:**
 - The system SHALL comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA to ensure it is usable by individuals with disabilities. This includes considerations for keyboard navigation, screen reader compatibility, sufficient color contrast, and proper semantic HTML.

OUTPUT







Registrar Dashboard | Judiciary In

+

localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar: syam

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

VIEW CASE DETAILS

CIN	Name of the defendant	Defendant address	Crime type	Date of crime	Place of crime	Name of arresting officer	Date of arrest	Posted by	Status
2025-FRAUD-298343	sreekanth	idupulapaya	fraud	2025-11-15	rkvalley	yuvaraj	2025-11-17	Raghuram	null

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Registrar Dashboard | Judiciary In

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localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar: syam

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

VIEW CASE DETAILS

CIN	Assigned	Assign
2025-FRAUD-168546	YES	Assigned
2025-FRAUD-298343	null	Assign
2025-FRAUD-595940	YES	Assigned
2025-FRAUD-873719	NO	Assign
2025-PROPERTY DISPUTE-251930	NO	Assign
2025-THEFT-133659	NO	Assign
2025-THEFT-575533	NO	Assign
2025-THEFT-734991	NO	Assign

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localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar: syam

Registrar Panel

View Case Details

Edit Case Details

Assign Case

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Announcements

View Assigned Cases

Logout

Registrar Dashboard

ASSIGN JUDGE TO CASE

Selected CIN:

2025-FRAUD-298343

Select Judge:

raju

Select Lawyer:

ram

Assign

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Summarize

Registrar: syam

Registrar Panel

View Case Details

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View Assigned Cases

Logout

Registrar Dashboard

VIEW CASE DETAILS

CIN	Assigned	Assign
2025-FRAUD-168546	YES	Assigned
2025-FRAUD-298343	YES	Assigned
2025-FRAUD-595940	YES	Assigned
2025-FRAUD-873719	NO	Assign
2025-PROPERTY DISPUTE-251930	NO	Assign
2025-THEFT-133659	NO	Assign
2025-THEFT-575533	NO	Assign
2025-THEFT-734991	NO	Assign

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Registrar Dashboard | Judiciary In

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localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar: syam

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

VIEW ASSIGNED CASE

CIN	Lawyer	Judge	Public Prosecutor
2025-FRAUD-168546	krish	raki	null
2025-FRAUD-298343	ram	raju	null
2025-FRAUD-595940	pinky	syam	null

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Registrar Dashboard | Judiciary In

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localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar: syam

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

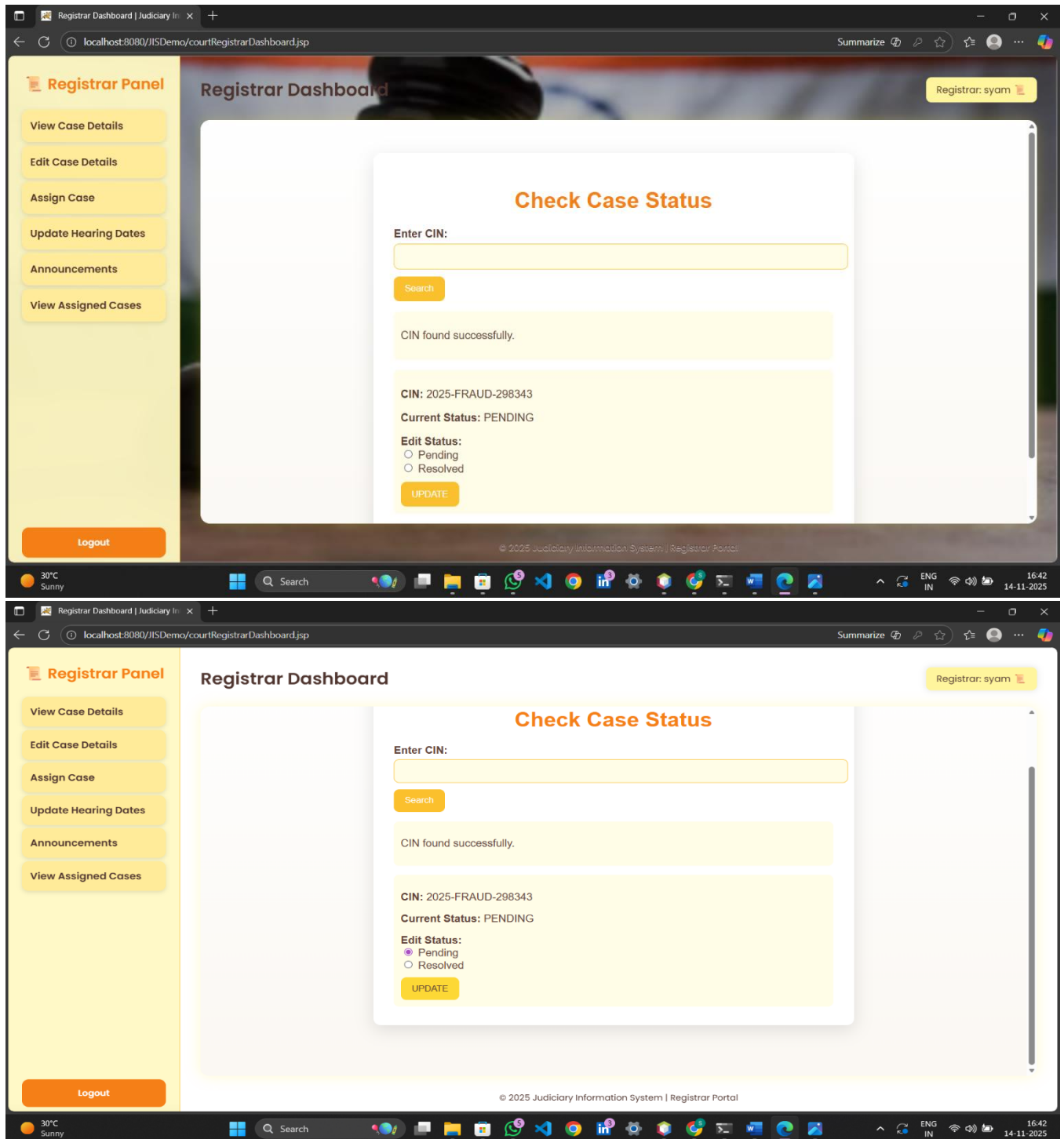
Check Case Status

Enter CIN:

Search

localhost:8080/JISDemo/editStatus.jsp

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Registrar Dashboard | Judiciary In

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localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar: syam

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

Check Case Status

Enter CIN:

Search

✓

Status updated to 'pending' for CIN: 2025-FRAUD-298343

CIN: 2025-FRAUD-298343
Current Status: pending
Edit Status:

Pending

Resolved

UPDATE

30°C
Sunny

Search

ENG
IN

16:42
14-11-2025

Registrar Dashboard | Judiciary In

+

localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar: syam

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

VIEW CASE DETAILS

CIN	Status	upload
2025-FRAUD-298343	pending	Edit
2025-FRAUD-595940	PENDING	Edit

30°C
Sunny

Search

ENG
IN

16:43
14-11-2025

Registrar Dashboard | Judiciary In

localhost:8080/JISDemo/courtRegistrarDashboard.jsp

Summarize

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

Registrar: syam

26	27	28	29	30	31	1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	1	2	3	4	5	6

Working day (click to assign)

Normal day

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Summarize

Registrar Panel

View Case Details

Edit Case Details

Assign Case

Update Hearing Dates

Announcements

View Assigned Cases

Logout

Registrar Dashboard

Registrar: syam

Check Case Status

Enter CIN:

Search

✔ Status updated to 'resolved' for CIN: 2025-FRAUD-298343

CIN: 2025-FRAUD-298343

Current Status: resolved

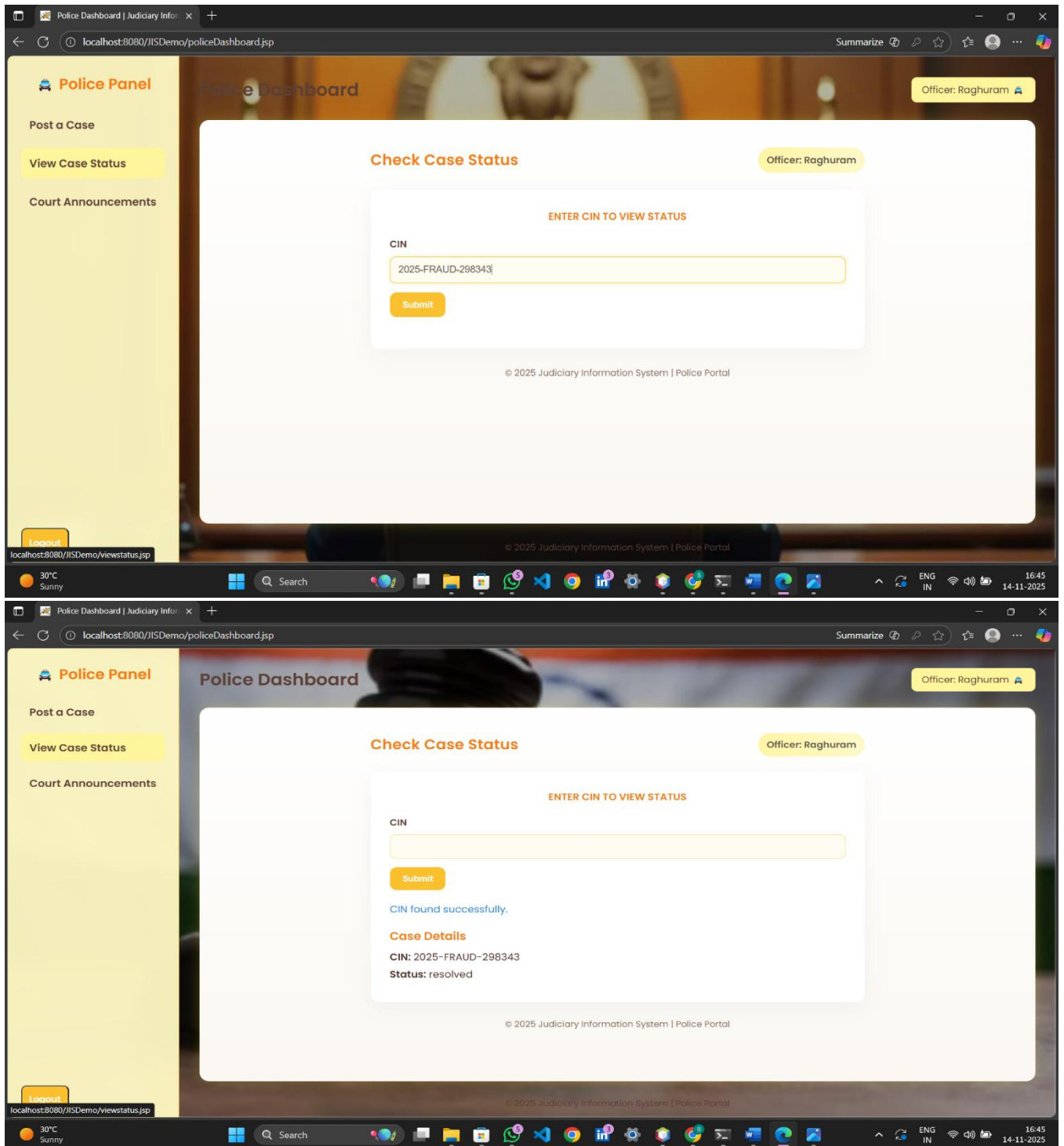
Edit Status:

Pending

Resolved

Update

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Activities

Google Chrome

Nov 14 16:29

Registrar x WhatsApp x Report-1. x Judge Da: x Judge Da: x Lawyer D: x Lawyer D: x Lawyer D: x

File /home/rguktrkvalley/lawyer_2.html#assigned

Lawyer Panel

View Assigned Cases

Access Previous Cases

Court Announcements

Logout

Lawyer Dashboard

Lawyer: Advocate A.K. Reddy

Assigned Cases

Case ID	Case Type	Parties	Next Hearing	Status
L2025-001	Theft	State vs Rajesh	2025-11-15	Active
L2025-002	Fraud	ABC Ltd vs Kumar	2025-11-20	Pending

Activities

Google Chrome

Nov 14 16:29

Registrar x WhatsApp x Report-1. x Judge Da: x Judge Da: x Lawyer D: x Lawyer D: x Lawyer D: x

File /home/rguktrkvalley/lawyer_2.html#previous

Lawyer Panel

View Assigned Cases

Access Previous Cases

Court Announcements

Logout

Lawyer Dashboard

Lawyer: Advocate A.K. Reddy

Access Previous Cases

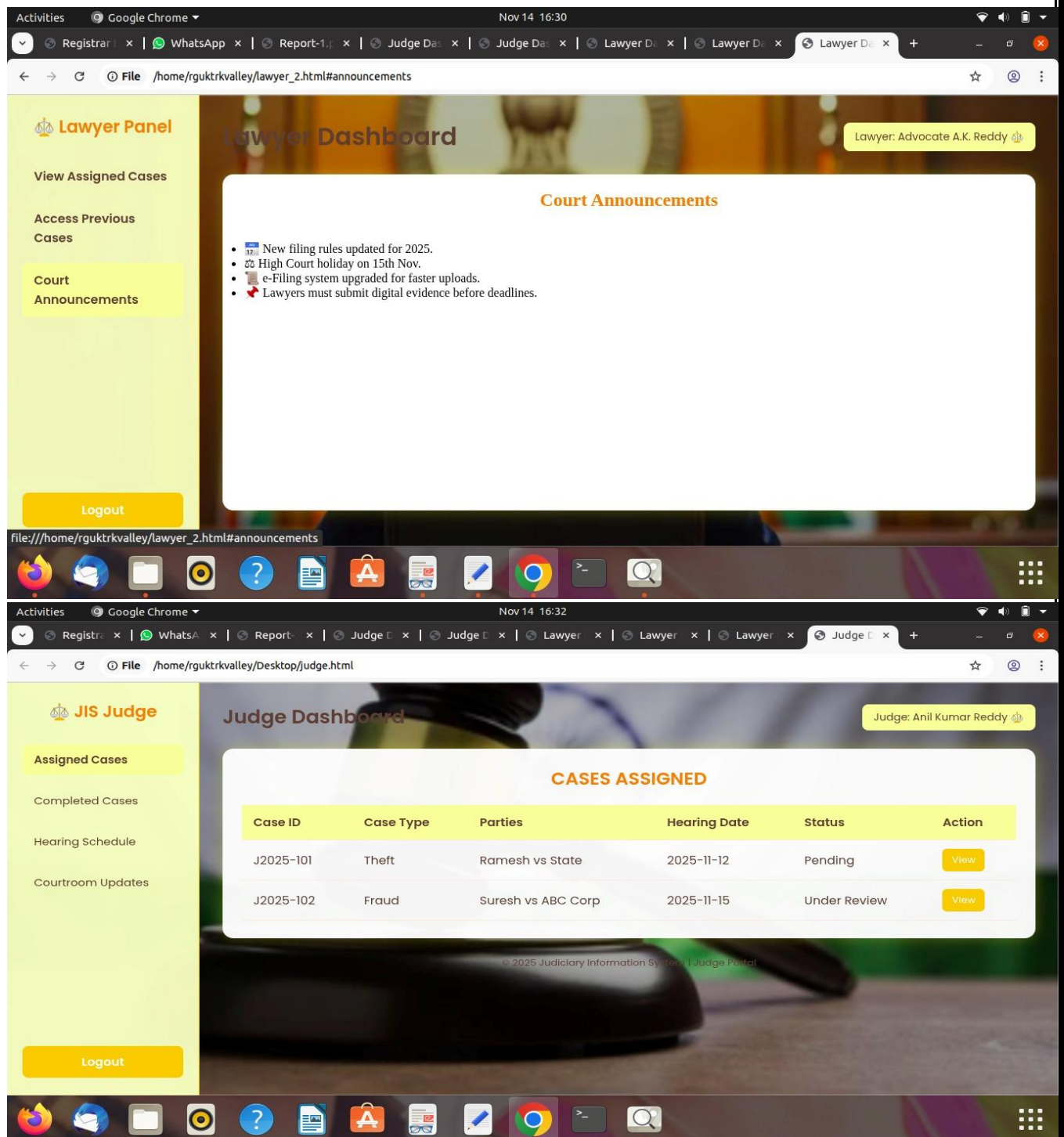
A small access fee applies for old case records.

Enter Case ID...

Case: L2024-014

Summary: Civil dispute | Verdict: In favour of plaintiff | Closed on 2024-08-12

49



JIS Judge

Assigned Cases

Completed Cases

Hearing Schedule

Courtroom Updates

Logout

Judge Dashboard

Judge: Anil Kumar Reddy

COMPLETED CASES

Case ID	Case Type	Parties	Judgment Date	Verdict
J2024-045	Property Dispute	Rahul vs Sunita	2024-12-18	In Favor of Rahul
J2024-102	Assault	State vs Dinesh	2025-01-22	Guilty

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JIS Judge

Assigned Cases

Completed Cases

Hearing Schedule

Courtroom Updates

Logout

Judge Dashboard

Judge: Anil Kumar Reddy

HEARING SCHEDULE

Date	Time	Case ID	Case Type	Parties
2025-11-12	10:00 AM	J2025-101	Theft	Ramesh vs State
2025-11-15	02:00 PM	J2025-102	Fraud	Suresh vs ABC Corp

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